

■ LIFE OS

# Command Centre

## Full Build Brief & Technical Specification

| Prepared For | Target Platform      | Date       | Version |
|--------------|----------------------|------------|---------|
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EXECUTIVE SUMMARY

This document captures the full design session in which LifeOS — a personal life-management web application — was conceived, designed, built, and deployed. It serves as the complete brief for rebuilding the application using Google Gemini models inside the Antigravity platform, with full mobile responsiveness.

## 1. ORIGIN & PURPOSE

The session began with a mentor-style framing: before taking action in life or business, one must first organise one's thoughts. The prescription was to build a personal operating system — a single web application giving full visibility across every life domain. The goal is clarity: knowing exactly where you stand in work, health, relationships, learning, and wealth at any given moment.

### MENTOR'S CORE PRINCIPLE

*"Show up with intention. Track your habits, advance one project, read one chapter, and call someone you love. Small edges, compounded daily, become extraordinary outcomes."*

## 2. THE 9 CORE MODULES

The application is organised into nine functional modules, each accessible via a persistent sidebar. All data is scoped to the individual user.

| Module           | Icon | Purpose                     | Key Features                                                                                |
|------------------|------|-----------------------------|---------------------------------------------------------------------------------------------|
| Work             | ■    | Track professional projects | Add/remove projects · Status: Active/Paused/Done · Priority (High/Med/Low) · Notes          |
| Habits           | ■    | Build periodic habit stack  | Cadences: Monthly · Quarterly · Yearly · Daily tick · Streak counter · Today % bar          |
| Skills           | ◆    | Map competency landscape    | 3 columns: Good At · Working On · Want to Learn · Tag chips · Add/remove freely             |
| Wealth           | ■    | Track assets & spending     | Investments: name, type, value, notes · Expenses: amount, category, date · Running £ totals |
| Learning         | ■    | Manage courses & knowledge  | In-progress with % slider · Platform field · Notes · Wishlist with priority tiers           |
| Reading          | ■    | Curate book pipeline        | Currently reading with % progress · Author field · Wishlist across 6 genres                 |
| Notes            | ■    | Capture & organise thoughts | Two-pane editor · Labels/categories · Filter by label · Auto-save · Created date            |
| Responsibilities | ■    | Map family commitments      | Add members · Fields: relationship, their expectations, how to improve, commitments         |
| Wish List        | ◆    | Track life goals & desires  | Add wishes · Tick when fulfilled · Pending/Fulfilled tabs · Progress bar (%)                |

## 3. DASHBOARD (OVERVIEW SCREEN)

The dashboard is the default landing screen. It aggregates live data from all 9 modules into a single glanceable view — assess your whole life in under 10 seconds.

### Stat Cards (7-grid)

One card per module. Shows key metric + label. Clicking navigates to that module.

### Learning Progress Panel

Top 3 in-progress courses with named progress bars.

### Reading Progress Panel

Top 3 in-progress books with named progress bars.

### Daily Intention

A mentor quote in a styled callout box. Sets the daily tone.

## 4. DESIGN SYSTEM & VISUAL LANGUAGE

### Colour Palette — Minimalist White / Black / Chocolate

| Token      | Hex     | Usage                                                   |
|------------|---------|---------------------------------------------------------|
| --white    | #FAFAF8 | Page background, card background                        |
| --off      | #F2EFE9 | Sidebar bg, secondary cards, stat card hover            |
| --cream    | #E8E2D9 | Borders, progress track, dividers                       |
| --choco-lt | #C8A882 | Dot pattern, icon accents, chip borders, streaks        |
| --choco    | #8B6343 | Primary buttons, active nav, active tab, progress fills |
| --choco-dk | #5C3D2E | Sidebar active state, section headers, deep CTA         |
| --ink      | #3A2516 | All body text, headings                                 |
| --muted    | #9C8B7A | Secondary text, placeholders, meta labels               |
| --success  | #4A7C59 | Done status, Good At skills column                      |
| --danger   | #A84242 | Delete / remove actions                                 |

■ Typography

- Headings / Section titles: DM Serif Display (serif, weight 400, italic variant for quotes)
- Body / UI text: DM Sans (weights 300, 400, 500, 600, 700)
- Base font size: 14px — scale from 11px captions to 34px hero title

■ 3D Dot Pattern Background (Signature Visual)

A repeating radial-gradient dot grid applied to sidebar and main at different densities, creating subtle 3D depth without any imagery.

Sidebar: radial-gradient(circle, rgba(139,99,67,.12) 1.2px, transparent 1.2px) / 20px 20px

Main bg: radial-gradient(circle, rgba(139,99,67,.07) 1px, transparent 1px) / 28px 28px

■ Component Library

| Component      | Specification                                                                              |
|----------------|--------------------------------------------------------------------------------------------|
| Card           | White bg, 14px radius, 1px --cream border, subtle shadow. Elevated variant: deeper shadow. |
| AddRow         | Flex row: text input (flex:1) + chocolate '+' button. Enter key triggers add.              |
| Tabs           | Pill buttons. Inactive: ghost border. Active: filled chocolate bg, white text.             |
| ProgressBar    | 5px track in --cream. Animated fill. Displays % value right-aligned.                       |
| XBtn           | 24x24px off-white square button with × glyph. Removes item immediately.                    |
| Chip/Tag       | Inline tag, coloured background tint + border. Used in Skills module.                      |
| Field/TextArea | 1.5px border, 10px radius. Focus: chocolate border + 3px glow ring.                        |
| NavBtn         | Full-width, 3px left border indicator when active, icon + label text.                      |

■ 5. LAYOUT ARCHITECTURE

■ Desktop (>700px)

Two-column flex layout. Left: fixed 220px sidebar (logo, nav list, footer). Right: scrollable main with centred 840px max-width panel, 36px padding, dot-bg.

■ Mobile (≤700px) — REQUIRED FOR REBUILD

MOBILE REQUIREMENTS

- Sidebar hidden off-screen (left: -220px), slides in on hamburger tap
- Hamburger button (■): fixed top-left, 38x38px chocolate, z-index 200
- Semi-transparent overlay closes sidebar when tapped outside
- Main padding reduced to 20px 16px on mobile
- Dashboard grid: 2 columns on mobile (not 4-5)
- 3-column grids (Wealth expenses) collapse to 2 columns
- Notes module: single column — list stacked above editor
- All touch targets minimum 44x44px (Apple HIG compliant)
- Progress sliders: height 28px on mobile for easy touch
- No font sizes below 13px on mobile
- No horizontal scroll anywhere
- Bottom nav bar option: 5 primary icons replacing sidebar on mobile
- Safe area insets: padding-bottom env(safe-area-inset-bottom) for iPhone notch

■ 6. DATA MODEL (JSON SCHEMA)

All state lives in one document per user. Original implementation: localStorage (key: 'lifeos-v6'). Antigravity rebuild: user-scoped backend store.

```
{ work: { projects: [{ id, name, status, description, priority, added } ] }, habits: { monthly:[], quarterly:[], yearly:[ ] }, // each habit: { id, name, ticks: { "YYYY-MM-DD": true } } skills: { goodAt:[], workingOn:[], wantToLearn:[ ] }, // each: { id, name } wealth: { investments:[], expenses:[ ] }, // investment: { id, name, type, value, notes } // expense: { id, name, amount, category, date } learning: { current:[], wishlist:[ ] }, // current: { id, name, platform, progress (0-100), notes } // wishlist: { id, name, priority } reading: { current:[], wishlist:{ } }, // current: { id, name, author, progress (0-100) } // wishlist keys: Business | Startup | Finance | // Geo-politics | Self-development | History notes: { items: [{ id, title, content, label, created } ] }, family: { members: [{ id, name, relationship, expectations, howToBeBetter, commitments } ] }, wishes: { items: [{ id, name, done, added, doneDate } ] } }
```

■ 7. KEY INTERACTIONS & BEHAVIOURS

| Interaction          | Behaviour                                                                      |
|----------------------|--------------------------------------------------------------------------------|
| Add item             | Type in AddRow + press Enter or click +. Item appended. Input clears.          |
| Delete item          | × button removes immediately. No confirmation required.                        |
| Tick habit           | Toggles ticks[today()]. Button fills chocolate + checkmark. Streak increments. |
| Cycle project status | Active → Paused → Done → Active. Border colour matches.                        |
| Progress slider      | HTML range input. onInput updates live. Bar animates (0.4s ease).              |
| Notes editor         | Click note in sidebar list to open right-pane editor. All fields auto-save.    |
| Family card          | Click member pill to open their 4-field card. Auto-saves on every input.       |
| Wish toggle          | Done items move to Fulfilled tab with strikethrough + fulfilled date.          |
| Dashboard nav        | Clicking any stat card calls goto(sectionId) to switch the active view.        |

■ 8. TECHNICAL IMPLEMENTATION (ORIGINAL BUILD)

■ Stack

- React 18 — UI framework via ES modules (esm.sh CDN, no build step needed)
- htm — JSX-alternative tagged template literals, no Babel compilation
- localStorage — data persistence (key: 'lifeos-v6')
- Single index.html — zero dependencies, deploys to any static host

- Vercel — static hosting, auto-deploy from GitHub (repo: smit-modi1/LifeOS)
- Live URL: <https://life-os-hazel-iota.vercel.app>

■ Key Bug Fixed During Build

First version used Babel Standalone to compile JSX in-browser. On large scripts (800+ lines) Babel silently crashed — buttons appeared but did nothing. Fix: switched to htm + React ES modules. Native browser execution, no compilation, 100% reliable. All 9 modules fully functional.

■ Deployment Steps (for reference)

- **Step 1:** Vercel project 'life-os' created under team smit-modis-projects-c6f07f9a
- **Step 2:** File uploaded as 'lifeos.html' — Vercel 404 (needed index.html)
- **Step 3:** Renamed to 'Index.html' (capital I) — Linux case-sensitive, still 404
- **Step 4:** Renamed to 'index.html' (lowercase) — resolved. App live.
- **Step 5:** Buttons not working — root cause: silent Babel crash on large file
- **Step 6:** Full rewrite using htm + ES modules. All buttons functional. Deployed.

■ 9. REBUILD SPECIFICATION — GEMINI IN ANTIGRAVITY

THIS IS THE ACTIVE BUILD TARGET

All sections above are the reference spec. This section defines how to rebuild LifeOS inside Antigravity using Gemini models, with a proper backend, auth, and AI-powered intelligence layer.

■ Architecture Recommendations

| Layer      | Recommendation                                                  |
|------------|-----------------------------------------------------------------|
| Frontend   | React (Vite) or Next.js with Antigravity component library      |
| State      | React Context + useReducer, or Zustand per module               |
| Backend/DB | Antigravity data layer — one document per user, keyed by userId |
| Auth       | Antigravity auth (Google/email SSO) — gate all routes           |
| AI         | Gemini 1.5 Flash (fast) or Pro (complex) via Antigravity SDK    |
| Hosting    | Antigravity deploy or Vercel + Antigravity backend              |

■ Gemini Integration Features

| Feature               | Where                   | What Gemini Does                                                                        |
|-----------------------|-------------------------|-----------------------------------------------------------------------------------------|
| Daily Briefing        | Dashboard load          | Reads all 9 modules, generates a 3-sentence personalised daily intention                |
| Habit Coach           | After ticking habits    | Analyses streak patterns, sends motivation or insight (e.g. 'You skip yoga on Fridays') |
| Reading Companion     | Reading Now — each book | 'Ask Gemini' button: Q&A; on book themes, summaries, reading notes                      |
| Weekly Review         | Dashboard or Notes      | Synthesises week's habit ticks, project moves, notes into structured review             |
| Smart Notes           | Notes editor            | Summarise long notes, extract action items, suggest which module idea belongs in        |
| Wealth Insights       | Wealth module           | Analyses investments + expenses, surfaces allocation or spending pattern observations   |
| Responsibility Nudges | Family module           | Based on expectations fields, periodically suggests specific relationship actions       |

■ Backend User Document — Additional Fields

- `userId`: string (from Antigravity auth)
- `createdAt` / `updatedAt`: ISO timestamps
- `lastActiveSection`: string — restore user to last module on return
- `preferences`: { `currency`: 'GBP', `theme`: 'light', `notifications`: boolean }

- `geminiHistory`: [] — store last N Gemini interactions per module for context

## ■ 10. BUILD CHECKLIST FOR ANTIGRAVITY

Use this as your task list when building in Antigravity:

### ■ SETUP

- ■ Antigravity project created with auth enabled
- ■ User data schema initialised per Section 6 + backend fields
- ■ Google Fonts loaded: DM Sans + DM Serif Display
- ■ CSS variables set per Section 4 colour palette
- ■ Dot-pattern backgrounds on sidebar and main
- ■ Mobile viewport meta tag in HTML head

### ■ NAVIGATION

- ■ Desktop: 220px fixed sidebar — logo, nav list with icons, footer
- ■ Mobile: hamburger + slide-in drawer + semi-transparent overlay
- ■ Active section: 3px left border + white background highlight
- ■ Stat cards on dashboard navigate to corresponding module

### ■ DASHBOARD

- ■ 7 stat cards in auto-fill grid (2-col mobile, 4-col desktop)
- ■ Learning progress bars for top 3 in-progress courses
- ■ Reading progress bars for top 3 in-progress books
- ■ Gemini daily intention (with static fallback while loading)

### ■ ALL 9 MODULES

- ■ Work: projects CRUD · status cycle · priority · notes textarea
- ■ Habits: 3 cadences · daily tick · streak counter · today completion %
- ■ Skills: 3 columns · chip tags · add/remove
- ■ Wealth: investments + expenses · running £ totals · date field
- ■ Learning: course progress slider + notes · wishlist with priority
- ■ Reading: book progress slider · wishlist by 6 genre categories
- ■ Notes: two-pane editor · labels · label filter · auto-save
- ■ Family: member selector pills · 4 textarea fields per member
- ■ Wishes: add · tick off · Pending/Fulfilled tabs · progress bar

### ■ GEMINI AI FEATURES

- ■ Daily briefing on dashboard — reads all module data
- ■ Habit coach message after ticking — stream response
- ■ Reading companion Q&A; per book
- ■ Weekly review generator
- ■ Smart notes: summarise / extract action items
- ■ Wealth insights commentary
- ■ Responsibility nudges per family member

### ■ MOBILE SPECIFIC

- ■ Bottom nav bar ≤768px (5 primary sections)

- ■ All touch targets minimum 44x44px
- ■ Dashboard: 2-col stat grid on mobile
- ■ No horizontal scroll on any screen
- ■ Notes: stacked single-column layout on mobile
- ■ Safe area insets for iPhone notch/home bar
- ■ Progress sliders: 28px height on mobile

## ■ QUALITY & POLISH

- ■ Data persists after page refresh (backend, not localStorage)
- ■ All buttons functional — no silent JS errors
- ■ Empty states shown for each module when no data
- ■ Auto-save on every input change (debounced 500ms)
- ■ Loading skeleton while Gemini responds
- ■ Error boundary for Gemini failures — graceful fallback

## ■ REFERENCE LINKS

- **Live App (Vercel):** <https://life-os-hazel-iota.vercel.app>
- **GitHub Repo:** <https://github.com/smit-modi1/LifeOS>
- **Vercel Project:** <https://vercel.com/smit-modis-projects-c6f07f9a/life-os>
- **DM Sans Font:** <https://fonts.google.com/specimen/DM+Sans>
- **DM Serif Display Font:** <https://fonts.google.com/specimen/DM+Serif+Display>
- **React via esm.sh:** <https://esm.sh/react@18>
- **htm Library:** <https://esm.sh/htm@3>

### CLOSING NOTE

*Every design decision — from the dot-pattern to the habit cadence structure — was arrived at iteratively. When rebuilding in Antigravity, treat this document as the source of truth. Deviate only where Gemini's capabilities allow you to go further. The spirit of the app is clarity, accountability, and compounding small daily edges into an extraordinary life.*